Gurmeher Singh Puri

M.Sc. Computer Science (Artificial Intelligence) — University of Freiburg

Freiburg im Breisgau, Germany · (+49) 179 9032502 · **Email:** gurmeher23@gmail.com

GitHub: github.com/Gurmeher23 · **LinkedIn:** gurmeher-singh-puri · **Portfolio:** gurmeher23.github.io

Available full-time from September 2026

PROFILE

Master's student in Computer Science (AI focus) at the University of Freiburg, completing a thesis at **Fraunhofer ISE** and the **Machine Learning Lab** (Prof. Dr. Frank Hutter, Dr. André Biedenkapp) on ML-driven optimisation of semiconductor laser processing. **Three years of production software engineering at Samsung R&D** (2× Employee of the Quarter). Seeking full-time roles in machine learning / ML engineering, software engineering, MLOps, or AI research.

MASTER'S THESIS & ACADEMIC RESEARCH

Automated Crack Detection & Process Optimisation in Laser-Assisted Splitting of SiC Wafers using Bayesian Optimisation — Feb 2026 – Aug 2026

Fraunhofer Institute for Solar Energy Systems (ISE) · Machine Learning Lab, University of Freiburg

Supervisors: Dr. André Biedenkapp & Prof. Dr. Frank Hutter

- **Why:** laser-assisted splitting of SiC wafers is a black-box process with 9 coupled laser parameters, where every trial recipe costs a wafer block and hours of measurement — built an ML + Bayesian Optimisation loop that replaces manual trial-and-error recipe tuning.
- **How:** turned raw laser-profilometer scans into an ML-ready dataset (2,160 blocks, 1,930 unique recipes, 68 engineered features); an unsupervised DBSCAN-based crack detector labels the data automatically, a stacked ML surrogate learns to predict crack quality from the 9 laser parameters, and **SMAC3**-driven Bayesian Optimisation searches for better recipes — closing the loop with the physical machine.
- **Models:** 6-model stacked surrogate — Random Forest, tuned XGBoost / LightGBM / CatBoost, and tabular foundation models (TabPFN, TabICL) under a Ridge meta-learner — lifting validated R^2 for cleavage cost from 0.52 (single-model baseline) to **0.85–0.92** and crack coverage to **0.93–0.95**.
- **Result:** multi-objective BO (SMAC3, ParEGO, 1,500 trials, coverage constraint) finds laser recipes reaching **85–95% crack coverage vs. 21–51% typical of manual tuning**, converging in ~100 physical experiments instead of ~10,000.

Master's Project — Electricity Carbon-Intensity Forecasting — Dec 2024 – Jul 2025

INATECH & Machine Learning Lab, University of Freiburg

GitHub: github.com/Gurmeher23/CO2-Emission-Intensity-Prediction-Germany-AutoML

- **Why:** Germany's growing share of renewables makes the grid's carbon intensity (gCO2/kWh) swing strongly by hour and season — accurate day-ahead forecasts enable carbon-aware computing and smarter energy management.
- **How:** built an **AutoML forecasting system** integrating hourly ENTSO-E data (generation, load, and cross-border flows for Germany + 11 neighbouring countries) with CO2map.de measurements; AutoGluon ensembles (LightGBM, CatBoost, XGBoost) compressed from multi-GB to **100–130 MB per model** for real-time deployment.
- **Result:** day-ahead forecasts for Germany nationally and all 16 federal states — **test R^2 0.93** (national, generation-based) and **0.91** (consumption-based) at ~8% SMAPE.

EMPLOYMENT HISTORY

Working Student — Machine Learning & Image Processing — Mar 2024 – Present

Fraunhofer ISE · Freiburg, Germany

- Core developer of **Spotlob**, an image-processing library for automated computer-vision detection of circles, lines, and edges on high-volume scientific imagery; applied to defect detection on solar cells.
- Designed modular, composable processing pipelines for feature extraction from large image datasets.
- Built supporting data infrastructure with **Apache Airflow** (orchestration), **MinIO** (S3-compatible storage), and **Flask** (webhook notifications) for reproducible experiment runs.

Software Engineer — Aug 2021 – Sep 2023

Samsung R&D Institute · Noida, India

★ 2× Samsung **Employee of the Quarter** (2022) · **Best Project Award** (RPA SWEET) · RPA **Best Team Award**

- Designed and shipped **12 internal tools and web services** (Python, JavaScript, RPA) automating engineering workflows — saving an aggregate **~149,000 engineering hours per year** (RPA SWEET ~120k · SWC Analyst ~20k · CR-CT ~9.2k).
- Owned **CI/CD pipelines** executing smoke tests on **1,000+ physical devices** for the Samsung Knox security platform.
- Authored automated test suites (Android Studio, JUnit) and managed distributed programming-contest infrastructure on DOMjudge.

Software Engineer Intern Jan 2021 – Jul 2021

Samsung R&D Institute · Noida, India

- Built two internal websites (JavaScript, Python, CSS) and contributed Samsung Knox automation test cases.

Web Developer Intern May 2019 – Jul 2019

Bharat Heavy Electricals Limited (BHEL) · India

- Developed the Student Tracker application using Eclipse and Tomcat 9.0.

TECHNICAL SKILLS

Machine Learning: Random Forest, Gradient Boosting (XGBoost, LightGBM, CatBoost), Gaussian Processes, Ridge/Lasso, stacked ensembles, AutoML (AutoGluon), Bayesian Optimisation (SMAC3, ParEGO), hyperparameter tuning, cross-validation, feature engineering

Deep & Foundation Models: TabPFN, TabICL, transformer-based tabular learners; PyTorch and Hugging Face workflows

Computer Vision & Signal Processing: OpenCV, RANSAC, DBSCAN, clustering, profilometer/image data, custom feature-extraction pipelines

NLP: NLTK, spaCy (Samsung research chatbot)

Programming: Python (expert), Java, C++, SQL, Bash/Shell, JavaScript

Data & Tooling: Pandas, NumPy, SciPy, scikit-learn, Jupyter, Matplotlib; Apache Airflow, MinIO, Docker, Git, GitLab CI/CD

Scientific Computing: HPC clusters (Helix, bwUniCluster), GPU computing, SLURM, remote SSH workflows

Languages: English — fluent (full professional proficiency) · **German** — B1, actively improving · **Hindi/Punjabi** — native

EDUCATION

M.Sc. Computer Science — Artificial Intelligence focus Oct 2023 – Sep 2026 (expected)

Albert-Ludwigs-Universität Freiburg · thesis defense August 2026

- Key courses: Foundations of AI, Foundations of Deep Learning, Deep Learning Lab, AutoML Lab, High-Performance Computing, Information Retrieval, Advanced Computer Graphics, Soft Robotics.

B.Tech. Computer Science and Engineering Jul 2017 – Jun 2021

Amity University Uttar Pradesh · Noida, India

SELECTED PROJECTS

Predicting Algorithm Runtime with TabPFN ***Python, TabPFN, Deep Learning***

Evaluated TabPFN v2 against the specialised deep model DistNet for algorithm runtime-distribution prediction (ML Lab, University of Freiburg).

Spotlob Image-Processing Library ***Python, OpenCV***

Open-source library at Fraunhofer ISE for reusable, composable feature-extraction pipelines on scientific imagery.

Chatbot for Research Assistance ***Python, NLTK, spaCy***

NLP assistant accelerating patent search and prior-art research at Samsung R&D.

Automation Process Tool ***Python, Airflow, MinIO, Docker, Flask***

Internal data-orchestration platform automating Samsung's testing and validation pipeline.

Automated Data Processing Tool ***Python, Pandas***

Cleansing and transformation of large-scale device-performance logs at Samsung R&D.

PUBLICATIONS

- "A Review on Cloud Computing" — **IEEE COMITCon 2019** (Int. Conference on Machine Learning, Big Data, Cloud and Parallel Computing), Amity University / Faridabad, India. ieeexplore.ieee.org/document/8776907
- "Cryptographic Technique for Communication Systems via the Diffie–Hellman Key Algorithm" (2021).
- "High Performance Computers and Computational Science" (2020).

ACHIEVEMENTS & CERTIFICATIONS

- Samsung **Employee of the Quarter** (Q1 & Q3 2022) and **Best Project Award** for the RPA SWEET Tool.
- Samsung RPA **Best Team Award**.
- Stanford Online: Algorithms — Design and Analysis.
- UiPath Certified Professional Automation Developer Associate.
- Microsoft Certification: Programming in HTML5 with JavaScript & CSS3.
- Google Digital Unlocked — Digital Marketing Expert.
- NSIC Entrepreneurship Orientation Program (EOP).